

PROJECT REPORT

For the fulfillment of:

CEA, Programming Fundamentals

Submitted to:

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ECAT Exam Portal

A Python Based Examination System

1. Introduction

This report describes the design and implementation of the ECAT Exam Portal, a desktop application built as a CEA project. The project was developed using the Python programming language along with the PyQt5 library for the graphical user interface. The purpose of this application is to simulate a real examination environment for ECAT preparation. It allows students to log in, take a timed multiple-choice exam, and immediately receive their result. An admin panel is also provided for managing the question bank and viewing student results.

2. Objectives

The main objectives of this project were:

- To build a fully working GUI-based exam system using Python and PyQt5.
- To implement a secure login system for both admin and student roles.
- To provide a live countdown timer during the exam.
- To calculate and display results with scoring, percentage, and grade.
- To give the admin tools to add, delete, and review questions and results.

3. Tools and Technologies Used

- **Python 3.14** - main programming language
- **PyQt5** - GUI framework for building the desktop interface
- **Qt Designer** - tool used to design the .ui layout file
- **QStackedWidget** - used to manage multiple pages in one window
- **QTimer** - used for the live 20-minute exam countdown
- **uic.loadUi()** - used to load the Qt Designer .ui file at runtime

4. System Structure

The project consists of three files that work together:

4.1 ecat.py

This file holds all the shared data and logic. It contains the question bank (a list of 10 dictionaries, each with an ID, subject, question, four choices, and the correct answer), the constants for marks (+4 correct, -1 wrong, 0 skip), the result storage list, and the `calculate_result()` function which grades the exam and returns a result dictionary.

4.2 Ecat.ui

This is an XML file created with Qt Designer that defines the entire visual layout of the application. It describes every widget (buttons, labels, text boxes, radio buttons) and their positions on screen. The application has one outer `QStackedWidget` with five main pages: Main Menu, Login, Student Menu, Exam, and Admin Menu. The Admin Menu contains an inner `QStackedWidget` with seven sub-pages for different admin functions.

4.3 main.py

This is the entry point of the application. It imports PyQt5 and the data from `ecat.py`, loads the `.ui` file using `uic.loadUi()`, and defines the `ECATApp` class which connects every button to its corresponding function. All navigation, login logic, exam flow, timer, and admin operations are handled here.

5. Features

5.1 Login System

The application has two separate login portals. The admin logs in with a fixed username and password. The student must enter a username, password, full name, and roll number. Both portals have a lockout mechanism after three failed attempts to prevent unauthorized access.

5.2 Exam Module

Once the student starts the exam, questions are shown one at a time with four radio button options. A 20-minute timer counts down at the top of the screen and turns red in the last five minutes. The student can answer and press Next, skip a question, or submit early. The exam auto-submits when the timer reaches zero.

5.3 Marking Scheme

The marking scheme follows standard ECAT rules:

- **Correct answer:** +4 marks
- **Wrong answer:** -1 mark (negative marking)
- **Skipped question:** 0 marks (no penalty)

5.4 Result and Grading

After the exam, the system calculates the total score, percentage, and assigns a grade. The grading criteria are:

- **80% and above:** EXCELLENT
- **65% to 79%:** GOOD
- **50% to 64%:** AVERAGE
- **Below 50%:** BELOW AVERAGE

5.5 Admin Panel

The admin panel provides the following tools:

- View all questions in the question bank
- Add a new question with subject, options, and correct answer
- Delete a question by its ID
- View question statistics broken down by subject
- View all student results in a summary list
- Search a detailed result for any student by roll number
- View class statistics including highest, lowest, average score, pass/fail count, and grade distribution

6. Question Bank

The application comes with 10 pre-loaded questions covering the following subjects: Biology (2 questions), Chemistry (2 questions), Physics (2 questions), English (1 question), History (1 question), General Knowledge (1 question), and Programming (1 question). Each question has four options (A, B, C, D) and one correct answer. New questions can be added and existing ones deleted by the admin at any time.

7. How the Application Works

When the application is launched, the main menu is shown with two buttons: Admin and Student. Selecting Student opens the login form which includes fields for username, password, full name, and roll number. After a successful login the student menu appears with options to start the exam or view the rules. When the exam starts, the first question is displayed along with the countdown timer. The student selects an answer using the radio buttons and clicks Next to proceed. They may also click Skip to move on without answering. On the last question the Next button is replaced by a Submit button. Once submitted, a popup message displays the final score, percentage, and grade.

The Admin portal is accessed from the main menu. After logging in, a sidebar menu on the left side lists all admin tools. Clicking any option loads the corresponding page in the content area on the right. All results collected during the session are available to the admin for review.

8. Challenges Faced

- Connecting the .ui file to Python code required understanding of how PyQt5 widget names map to object attributes in the ECAT App class.
- Managing two separate QStackedWidgets (one outer and one inner inside the admin page) required careful tracking of page indices.
- Properly unchecking radio buttons when loading a new question without causing mutual-exclusivity errors required temporarily setting `setAutoExclusive(False)`.
- Making the exam timer stop correctly in all cases (normal submit, skip on last question, time up) required careful placement of the timer stop call.

9. Conclusion

The ECAT Exam Portal is a complete, working desktop application that successfully demonstrates the use of Python and PyQt5 for building real-world software. The project covers important programming concepts including object-oriented programming, event-driven GUI development, data management using lists and dictionaries, and modular code organization across multiple files. The application fulfills all the objectives set for the CEA project. It provides a functional exam portal with role-based access, a timed exam engine, automatic grading, and full admin control over the question bank and results.

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